Joint Probabilities : $P(A,B) = P(B,A)$

basically

$P(A \cap B) = P(B \cap A)$

Conditional Probabilities

Bayes' Rule

$$\frac{\text{likelihood} \quad \text{prior}}{\text{evidence}} \quad \frac{p(B|A)P(A)}{p(B)}$$

$$P(A|B)p(B) = p(B|A)p(A)$$

posterior · evidence · likelihood · prior

$$P(B) = P(A)P(B|A) + P(\sim A)P(B|\sim A)$$

or

$$P(B) = \sum_a P(A=a)P(B|A=a)$$

* Finding Distribution for the each class's likelihood is difficult.

we can multiply because we know their independent of eachother

Likelihood

$$p(x_n | y_n = k) = \prod_M p(x_{nm} | y = k)$$

Assuming a PDFs like a Normal Distribution for every predicto x in x_n

$P(x_{n0} | y_n = k) \quad \cdots \quad P(x_{nM} | y_n = k)$

in-sample data's # occurence of class k / in-sample data size

posterior

$$p(y_n = k | x_n) = \frac{\text{Prior} \quad \text{Likelihood}}{\text{Evidence}} \quad \frac{p(y_n = k) \prod_M p(x_{nm} | y = k)}{\sum_{i=1}^{k} p(y_n = i)\, p(x_n | y_n = i)}$$

Assuming PDF for predictor in x_n for every class k

$P(x_{n0} | y_n = 0) \quad \cdots \quad P(x_{nM} | y_n = 0)$

$P(x_{n0} | y_n = 0) \quad \cdots \quad P(x_{nM} | y_n = k)$

Maximum Likelihood

Similar to Logistic Regression
- minimize the log likelihood
- likelihood found via PDF

Log-likelihood ratio

Is class k

$$p(y_n = k | x_n) = \frac{p(y_n = k) \prod_M p(x_{nm} | y_n = k)}{p(x_n)}$$

Is not class k

$$p(y_n \neq k | x_n) = \frac{p(y_n \neq k) \prod_M p(x_{nm} | y_n \neq k)}{p(x_n)}$$

$$\ln\left(\frac{p(y_n = k | x_n)}{p(y_n \neq k | x_n)}\right) = \ln\left(\frac{\frac{p(y_n = k) \prod_M p(x_{nm} | y_n = k)}{p(x_n)}}{\frac{p(y_n \neq k) \prod_M p(x_{nm} | y_n \neq k)}{p(x_n)}}\right)$$

$$= \ln\left(\frac{p(y_n = k) \prod_M p(x_{nm} | y_n = k)}{p(y_n \neq k) \prod_M p(x_{nm} | y_n \neq k)}\right)$$

$$= \ln\left(\frac{p(y_n = k)}{p(y_n \neq k)}\right) + \ln\left(\frac{\prod_M p(x_{nm} | y_n = k)}{\prod_M p(x_{nm} | y_n \neq k)}\right)$$

$$= \ln\left(\frac{p(y = k)}{p(y \neq k)}\right) + \sum_M \ln\left(\frac{p(x_{nm} | y_n = k)}{p(x_{nm} | y_n \neq k)}\right) > 0$$

$y_n \to k$

$$\ln\left(\frac{P(A|D_7)}{P(B|D_7)}\right) = \ln\left(\frac{P(A)}{P(B)}\right) + \sum_M \ln\left(\frac{P(w_m|A)}{P(w_m|B)}\right)$$

$$= \ln\left(\frac{2/3}{1/3}\right) + \ln\left(\frac{P(w_0|A)}{P(w_0|B)}\right) + \ldots + \ln\left(\frac{P(w_0|A)}{P(w_0|B)}\right)$$

$$\ln(2) + \ln\left(\frac{1/4}{1/6}\right) +$$

$D = \{ w2\ w2\ w3\ w4\ w5\ w5 \}$

$$\sum_m^D \ln\left(\frac{P(w_2|A)}{P(w_2|B)}\right) + \ln\left(\frac{P(w_2|A)}{P(w_2|B)}\right) \ldots$$

Use Word Count as Shorthand

$$\ln\left(\frac{P(A)}{P(B)}\right) + \sum_m^M \text{wordCount}(w_m) \ln\left(\frac{P(w_m|A)}{P(w_m|B)}\right)$$

A naive Bayes classification example

- Consider the document collection represented in the table
- Use log-likelihood ratios to classify of the unlabeled document

Document	Category	Word-1	Word-2	Word-3	Word-4	Word-5	Total Words
D1	A	3	1	2	2	1	9
D2	A	2	0	3	2	0	7
D3	B	2	3	1	2	2	10
D4	A	1	0	2	3	1	7
D5	B	1	3	1	1	2	8
D6	A	2	1	1	3	2	9
D7	?	0	2	1	1	2	6

A naive Bayes classification example

- Consider the documents in category A, from which we can estimate conditional probabilities of the form $P(w_m | C_n = A)$

Document	Category	Word-1	Word-2	Word-3	Word-4	Word-5	Total Words
D1	A	3	1	2	2	1	9
D2	A	2	0	3	2	0	7
D4	A	1	0	2	3	1	7
D6	A	2	1	1	3	2	9
Total Counts		8	2	8	10	4	32
p(w_i|A)		1/4	1/16	1/4	5/16	1/8	-

$$P(\text{category} = A) = \frac{4}{6}$$

$$P(\text{category} = B) = \frac{2}{6}$$

$$\prod P(w_i | A) = P(w_1 | A)P(w_2 | A) \ldots P(w_5 | A)$$

$$\ln\left(\prod P(w_i | A)\right) = \ln(P(w_1 | A)) + \ln(P(w_2 | A))$$

$$P(w_1 | B) = \frac{3}{18}$$

$$P(w_2 | B) = \frac{6}{18}$$

$$P(w_3 | B) = \frac{5}{18}$$

$$P(w_4 | B) = \frac{1}{9}$$

$$P(w_5 | B) = \frac{1}{1}$$

$$P(w_6 | B) = \frac{2}{9}$$

$$\ln\left(\frac{p(A|B)}{p(B|A)}\right) + \sum \ln\left(\frac{p(w_{2m}|A)}{p(w_{2m}|B)}\right)$$

$$\ln(2) + \left(\ln\left(\frac{1/4}{1/6}\right)\right.$$

A naive Bayes classification example

- These are the resulting model parameters

Category	p(c)	$p(w_1\|c)$	$p(w_2\|c)$	$p(w_3\|c)$	$p(w_4\|c)$	$p(w_5\|c)$
A	2/3	1/4	1/16	1/4	5/16	1/8
B	1/3	1/6	1/3	1/9	1/6	2/9

- Which can be used to determine the category of document D_7

Document	Category	Word-1	Word-2	Word-3	Word-4	Word-5	Total Words
D7	?	0	2	1	1	2	6

$$\ln\left(\frac{P(A|D_7)}{P(B|D_7)}\right) = \ln\left(\frac{P(A)}{P(B)}\right) + \sum_M \ln\left(\frac{P(w_m|A)}{P(w_m|B)}\right)$$

$$= \ln\left(\frac{2/3}{1/3}\right) + \ln\left(\frac{P(w_0|A)}{P(w_0|B)}\right) + \ldots + \ln\left(\frac{P(w_0|A)}{P(w_0|B)}\right)$$

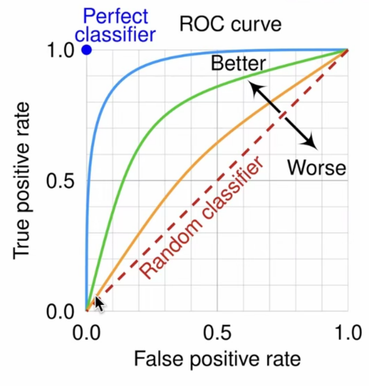
$$\ln(2) + \ln\left(\frac{1/4}{1/6}\right) +$$

$D = \{ w2\ w2\ w3\ w4\ w5\ w5 \}$

$$\sum_m^D \ln\left(\frac{P(w_2|A)}{P(w_2|B)}\right) + \ln\left(\frac{P(w_2|A)}{P(w_2|B)}\right) \ldots$$

Use Word Count as Shorthand

$$\ln\left(\frac{P(A)}{P(B)}\right) + \sum_m^M \text{wordCount}(w_m) \ln\left(\frac{P(w_m|A)}{P(w_m|B)}\right)$$

Classification metrics…

- Receiver Operating Characteristic (ROC) curve:
  - Graphic representation of a binary classifier performance
  - True Positive Rate (TPR) vs False Positive Rate (FPR)
- Area Under the Curve (AUC)
  - Perfect classifier: AUC = 1.0
  - Random classifier: AUC = 0.5